

Urban heat island in Southwark

The urban heat island is the extra heat a city holds compared with the countryside around it. Roads, pavements and buildings absorb heat by day and trap it (London Climate Ready Partnership, n.d.). Southwark sits at the centre of London's heat island, with summer temperatures estimated around 4.5°C above rural areas (Southwark Council, 2025).

Evidence

- central London was up to 10°C warmer than rural areas in the 2003 heatwave, and the heat island can add 5 to 6°C to night-time temperatures (London Climate Ready Partnership, n.d.)
- in the July 2022 heatwave the heat island averaged 2.3°C and peaked at 7.2°C, strongest at night and in central London (Simpson, Brousse and Heaviside, 2024)
- it caused an estimated 141 deaths in Greater London over that heatwave, 38% of all heat-related deaths (Simpson, Brousse and Heaviside, 2024)
- young children, Asian and Black residents and lower-income groups in London face surface temperatures up to 4°C hotter than wealthier, mainly White populations (Krenz and Amann, 2025)
- hot streets show 6 to 9% lower walking accessibility, limiting movement for the most exposed groups (Krenz and Amann, 2025)
- higher heat risk in London correlates directly with greater socio-economic vulnerability (Greater London Authority and Arup, 2024)
- green cover sets the local pattern: Kilburn and South Hampstead (38% vegetation) ran over 7°C hotter than nearby Regent's Park (89%) (Arup, 2023)

UHeatRank map

Arup's UHeatRank, produced for the GLA, ranks every LSOA in London from 1 (coolest) to 5 (hottest) using the UHeat tool (Greater London Authority and Arup, 2024). It models neighbourhood temperatures, which is closer to what people feel than surface temperature alone.

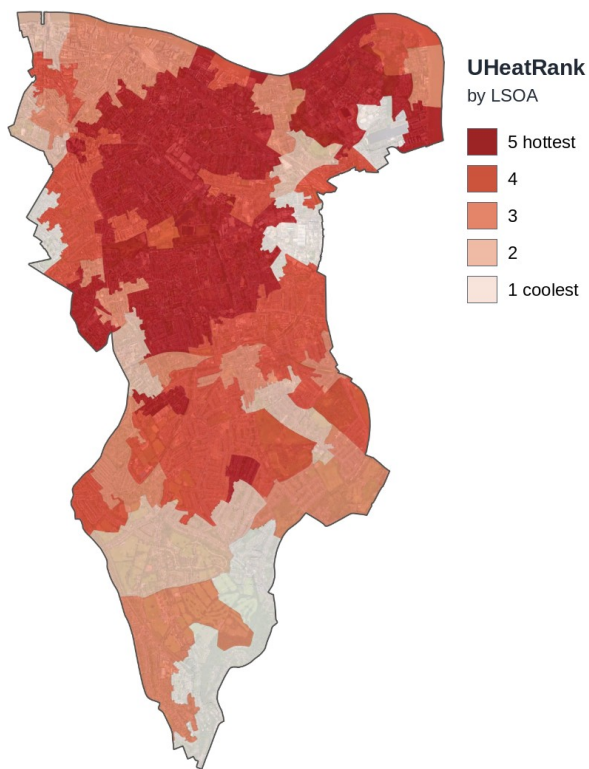


Figure 1. UHeatRank by LSOA, Southwark. Source: Greater London Authority and Arup (2024).

The hottest areas run through the north and centre of the borough. Walworth, Camberwell, Peckham and South Bermondsey combine the highest heat rank with the least green space and the highest deprivation (Southwark Council, 2025). The south of the borough and the larger parks rank lowest.

Land surface temperature

Satellite surface temperature shows the same pattern. Hot-season surface temperatures across south London range from about 33°C to 43°C, with the Thames and large green spaces coolest (WRI, 2026). Surface temperature exceeds air temperature, so this shows pattern, not rank.

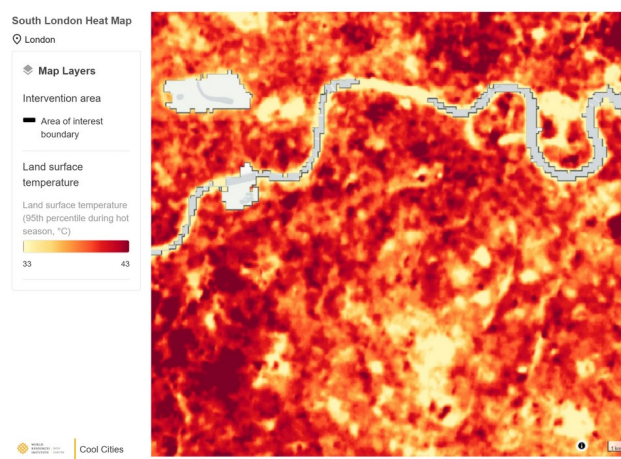


Figure 2. Land surface temperature, 95th percentile hot season (°C). Source: WRI (2026).

Works cited

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